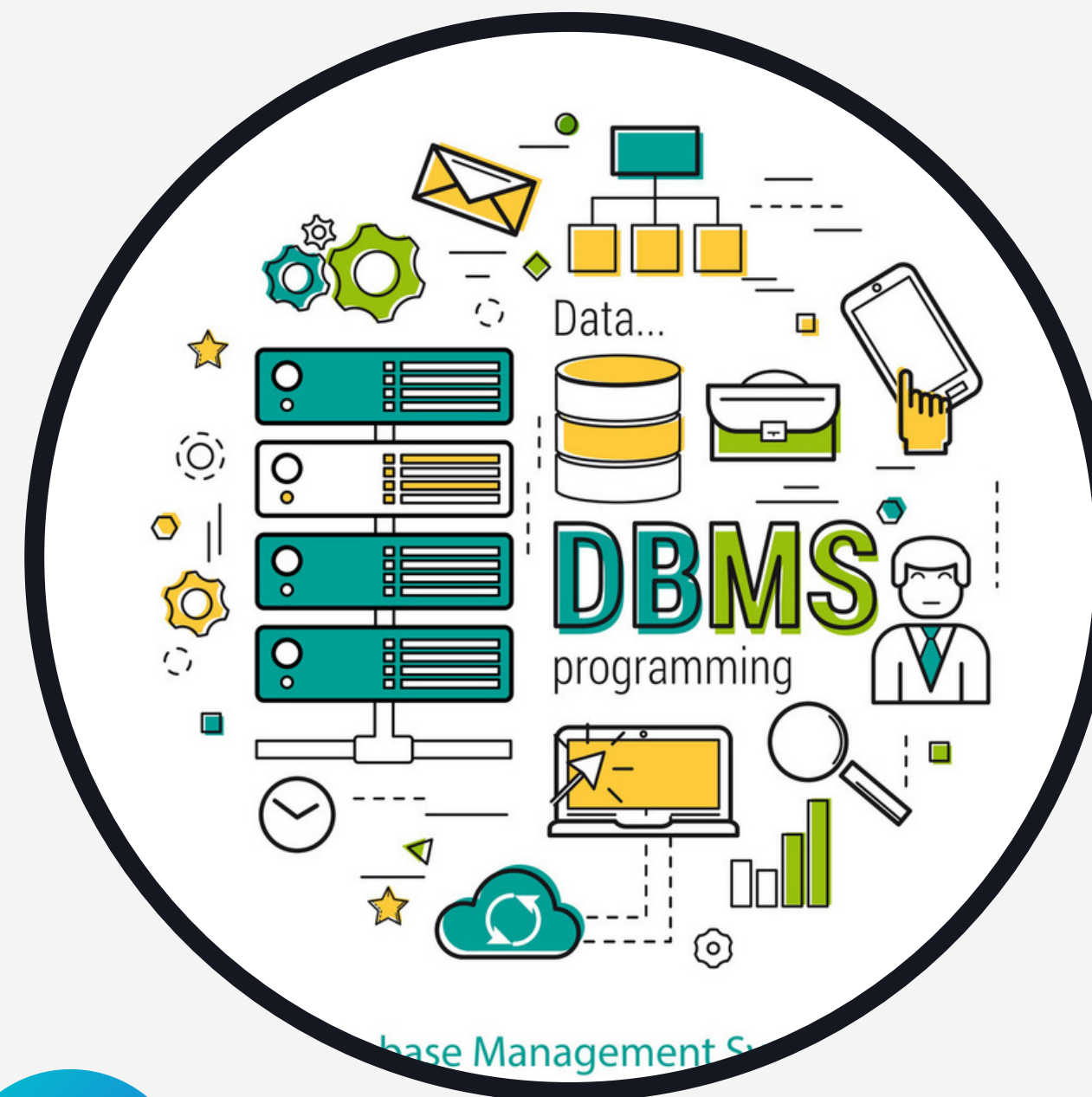


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DBMS V/S FILE SYSTEM

In the file system, there were various anomalies like insertion, deletion and updation anomalies i.e., if we insert, delete, or update data at one place then its change is not reflected in other places. But, In DBMS, it does not happen.

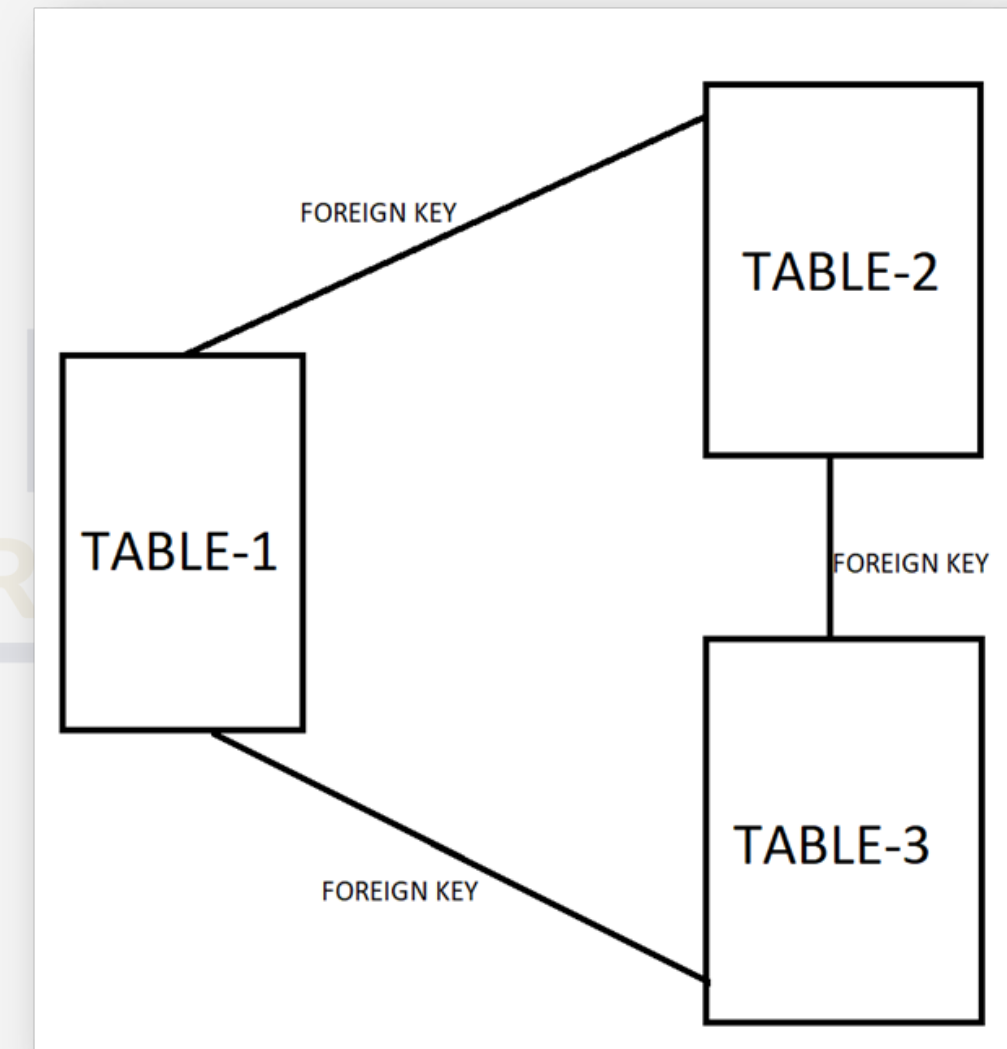
File system cannot handle large amount of data.

We cannot write complex queries in file system. But In DBMS, we can write queries.

File system is less secure. While, DBMS is more secure.

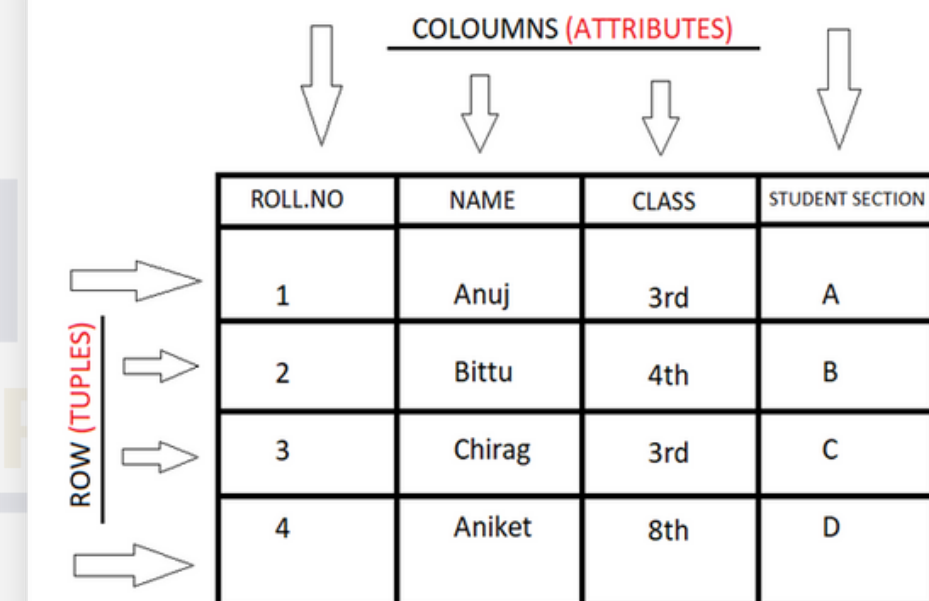
DBMS V/S RDBMS

In DBMS, data can be stored in any form, while in **RDBMS** stands for **Relational Database Management System**, here we store data in the form of tables which are related to each other and connected with the help of **Foreign Keys**.



WHAT IS TABLE?

A Table is a collection of rows and columns,
Rows are called **Tuples**.
Columns are called **Attributes**.



The diagram shows a table with four columns and four rows. Above the table, the text 'COLOUMNS (ATTRIBUTES)' is written in red, with four arrows pointing down to each column header. To the left of the table, the text 'ROW (TUPLES)' is written in red, with four arrows pointing right to each row. The table itself has the following data:

ROLL.NO	NAME	CLASS	STUDENT SECTION
1	Anuj	3rd	A
2	Bittu	4th	B
3	Chirag	3rd	C
4	Aniket	8th	D

KEY / SUPER KEY

It is an attribute or set of attributes which uniquely identifies tuples in a table .

There can be multiple keys in a table .

STUDENT'S TABLE				
ROLL.NO	NAME	COURSE	E-MAIL	CONTACT
1	Gaurav	C1	X	716
2	Vaibhav	C2	Y	820
3	Saurabh	C3	Z	905
4	Gaurav	C3	P	616
5	Saurabh	C1	Q	763



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CANDIDATE KEY

Candidate Key is a key with no redundant attributes. In other words, we can say that a key is said to be a candidate key if its any proper subset is not a super key itself.

There can be multiple candidate keys in a table.

Roll no. is the Candidate Key because it does not contain any redundant attribute but roll.no,name is not a Candidate Key.

ROLL.NO	NAME	COURSE	E-MAIL	CONTACT
1	Gaurav	C1	X	716
2	Vaibhav	C2	Y	820
3	Saurabh	C3	Z	905
4	Gaurav	C3	P	616
5	Saurabh	C1	Q	763

PRIMARY KEY

Primary Key is an attribute or set of attributes which uniquely identifies Tuples in a table.

Actually, there can be multiple candidate keys in a table out of which one key is selected by the Database Administrator (DBA) and assigned as the primary key of the table .

- There can only be ONE Primary Key in a table .
- Primary key cannot be NULL .

UNIQUE KEY

Unique Key is the candidate key of the table which is not assigned as primary key .

- It can be Multiple .
- It can be NULL .

<u>PRIMARY KEY</u>	<u>UNIQUE KEY</u>
Out of multiple candidate key, one special key is selected by Database Administrator called PRIMARY KEY . It is only one in the table.	It is the candidate key in the table which is not assigned as the PRIMARY KEY of the table . It can be multiple in a table . It can be NULL .

FOREIGN KEY

It is an attribute or set of attributes of one table that points to the primary key of another table.



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FUNCTIONAL DEPENDENCY

If there is a functional dependency from X to Y, then we say if we know the value of X, then can search the value of Y in the table.

ID --> NAME ✓

NAME --> ID ✗

CITY --> L-NAME ✗

L-NAME --> RELIGION ✓

RELIGION --> ID ✗

ID	F-NAME	CITY	CONTACT	L- NAME	RELIGION
1	GAURAV	BHOPAL	892	TIWARI	HINDU
2	VAIBHAV	INDORE	673	MISHRA	HINDU
3	SHADAB	INDORE	512	KHAN	MUSLIM
4	AMAN	GWALIOR	726	CLAUDIOUS	CHRISTIAN

PROPERTIES OF FUNCTIONAL DEPENDENCY

DBMS NOTES

ARMSTRONG AXIOMS :

1. If $X \twoheadrightarrow Y$
 $Y \twoheadrightarrow X$ ❌ $X \twoheadrightarrow X$ (is always true)

2. If $X \twoheadrightarrow Y$
 $A \twoheadrightarrow B$

$X, A \twoheadrightarrow Y, B$ ✅

3. If $X \twoheadrightarrow A, B, C$

$X \twoheadrightarrow A$
 $X \twoheadrightarrow B$
 $X \twoheadrightarrow C$ ✅

4. If $X, Y, Z \twoheadrightarrow A$

$X \twoheadrightarrow A$
 $Y \twoheadrightarrow A$ ❌
 $Z \twoheadrightarrow A$

5. If $A \twoheadrightarrow B$ and $B \twoheadrightarrow C$
 $A \twoheadrightarrow C$ (Transient Condition) ✅

PROPERTIES OF FUNCTIONAL DEPENDENCY

DBMS NOTES

PROPER SUBSET :

$S = \{a, b, c\}$

Proper Subset = $\{ \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{\phi\} \}$

$S = \{a, b\}$

Proper Subset = $\{ \{a\}, \{b\}, \{\phi\} \}$

PROPERTIES OF FUNCTIONAL DEPENDENCY

DBMS NOTES

ATTRIBUTE CLOSURE :

It is the can set of all attributes which are reachable from a particular attribute .

R(id, Fname, Lname, Caste, City, Pincode)

Id+ = {id, Fname, Lname, Caste, City, Pincode}

Fname + = {Fname}

Lname + = { Lname}

Caste + = {Caste}

City + = {City}

Pincode + = {City,Pincode}



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PROPERTIES OF FUNCTIONAL DEPENDENCY

DBMS NOTES

PRIME ATTRIBUTES :

The attributes which are part of a candidate key is called Prime Attributes .

NON - PRIME ATTRIBUTES :

The attributes which are NOT part of a candidate key is called Non - Prime Attributes .

NORMALIZATION is a technique to reduce the redundancy in the table.
REDUNDANCY means duplication of Data.

There are FOUR types of Normal Forms:

- 1 NF (1st Normal Form)
- 2 NF (2nd Normal Form)
- 3 NF (3rd Normal Form)
- BCNF (Boyce Codd Normal Form)

As we go to higher Normal Form, redundancy of data decreases.

Disadvantage : As we go to higher Normal Form, the query start to become complex.

1ST NORMAL FORM (1 NF).



DBMS NOTES

A table is said to be in 1 NF, if it does not contain any composite or multivalued attribute .

1. Composite Attributes :

An attribute is said to be an composite attribute if it is a combination of 2 or more attributes.

Example - Name, date of birth, Address

In order to avoid composite attributes, we add Extra Coloumns .

2. Multi-valued Attributes :

These are the attributes which can have more than 1 value at a time .

Example - Contact Number

To avoid multi-valued attributes, we add Extra Rows.

2ND NORMAL FORM (2 NF).

DBMS NOTES

- It should be in 1 NF .
- It should not have any partial dependency .

Partial Dependency : if any proper subset of candidate key yield any non-prime attribute then we say Partial Dependency exist .

3RD NORMAL FORM (3 NF).

DBMS NOTES

- It should be in 2NF .
- It should not have any Transitive Dependency .

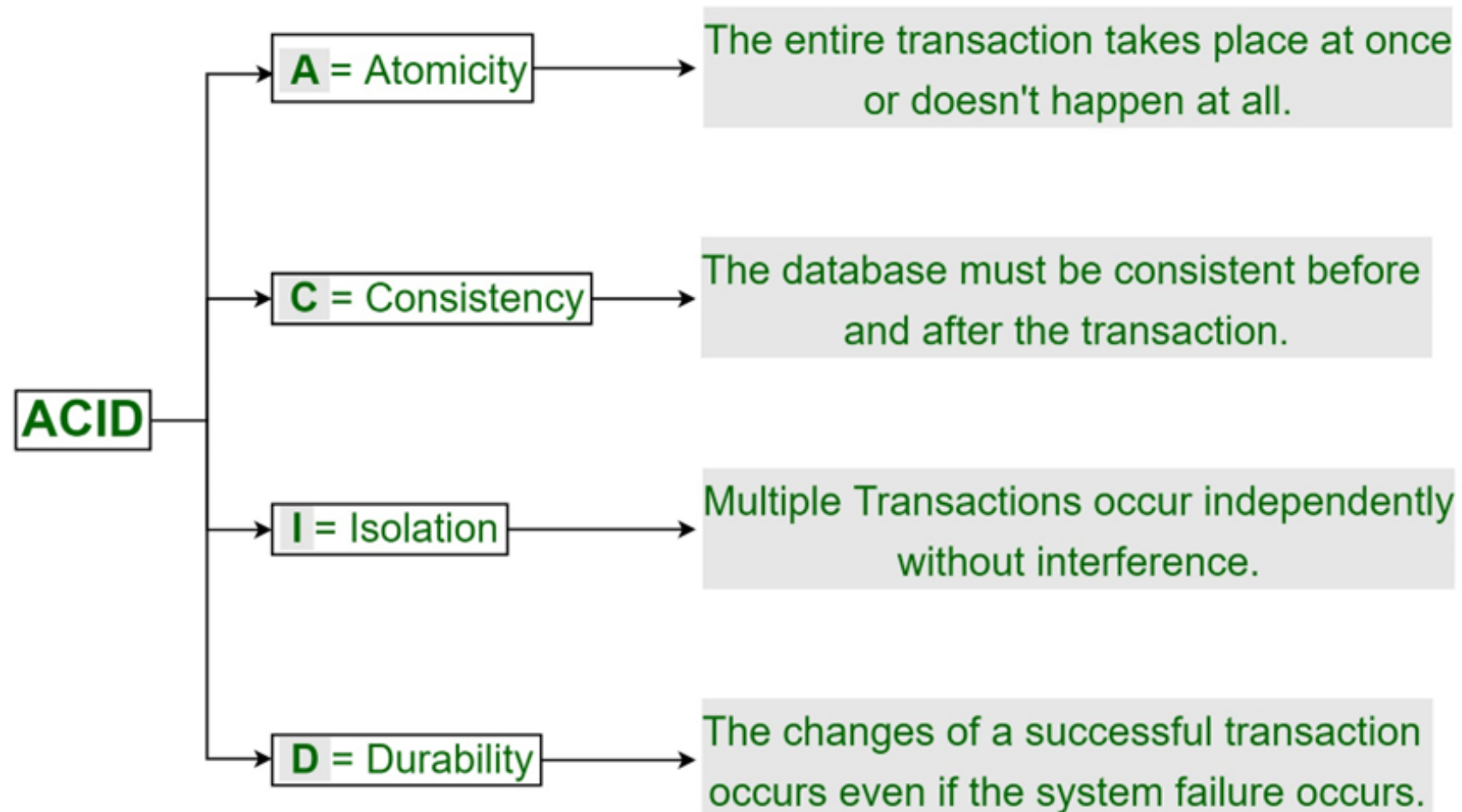
Transitive Dependency: If any non-prime attribute yield any other non-prime attribute then we say Transitive Dependency exists .

BOYCE CODD NORMAL FORM (BCNF)

DBMS NOTES

- It should be in 3NF .
- For every non-trivial functional dependency, left hand side must be a super key itself .

ACID Properties in DBMS



ABOUT SAI CRT:

DBMS NOTES

Sai Campus Recruitment Training is the best institute in Central India for Campus Placements located in Indrapuri, Bhopal. CRT stands for Campus Recruitment Training batch. This is a program where we teach students everything from basics to advance required to get placed in world's top MNC's like Amazon, TCS, Infosys, SAP, Capgemini, Accenture, Cognizant etc. After joining this batch you don't have to worry about anything, We cover every single thing required for your smooth selection. 2000+ students are already placed in their dream MNC's....., YES! you are next!

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ABOUT GAURAV TIWARI SIR:



DBMS NOTES

Mr. Gaurav Tiwari, a multi-faceted individual with a diverse range of talents and achievements. Mr. Tiwari is widely recognized as a prominent programming trainer, aptitude and logical reasoning instructor, and a proficient communicator. As a famous educator and emerging entrepreneur, he has left an indelible mark in the world of education and business.

Beyond his expertise in training, Mr. Gaurav Tiwari's passion for chess has led him to achieve international recognition as a rated chess player. He has represented himself in numerous national and international chess events, showcasing his strategic acumen and dedication to the game.

As a testament to his innovative abilities, Mr. Tiwari led the team "Ingenious," which emerged victorious in the prestigious Smart India Hackathon 2017, a platform where young minds collaborate to solve real-world challenges through technology.

His outstanding achievements have not gone unnoticed, as he was honored by the Governor of Madhya Pradesh State for securing the highest number of job offers from top multinational companies. This accolade exemplifies his exceptional skills and dedication in the corporate realm.



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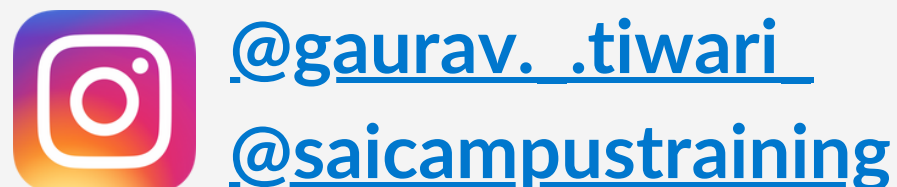
DBMS NOTES

Apart from his educational pursuits and entrepreneurial ventures, Mr. Gaurav Tiwari shares his experiences and knowledge with the world as a daily vlogger on his YouTube channel "Gauraverse." Through this platform, he has inspired and motivated thousands of viewers.

In a groundbreaking move, Mr. Tiwari established "Cryptovilla Cafe," India's first cryptocurrency-based cafe, demonstrating his forward-thinking approach to business and technology.

Moreover, his passion for coding and love for teaching have driven him to mentor over 10,000 students, making a lasting impact on the lives and careers of aspiring coders.

Mr. Gaurav Tiwari's remarkable journey and versatile talents make him an extraordinary individual, leaving an indelible mark in the domains of education, entrepreneurship, chess, and technology.



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